

Master's Thesis

A Data-Driven Analysis of the Impact of Online Reviews on Consumer Demand: Yelp and Amazon

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Abstract

This master's thesis investigates the relationship between online customer reviews and consumer demand using real-time datasets provided by Yelp and Amazon. More specifically, the study aims to analyze how factors like review lengths, review sentiments and review star ratings can affect business and product popularity and consumer demand. To achieve this, we did exploratory data analysis (EDA), Ordinary Least Squares (OLS) regression and random forest machine learning models for predictions in order to understand the relationship. The findings from the study show that online reviews do have a measurable amount of effect on consumer demand although the strength of the relationship depends on the platform. Overall, this thesis shows that online reviews can provide useful insights into consumer behavior and business popularity, while also showing the limitations of using review data alone to measure true consumer demand. This topic is relevant for businesses and product sellers that want to better understand how online customer feedback influences consumer behavior and business popularity in an increasingly digital marketplace.

The Python source code for this thesis as well as sample data sources can be found here: <https://github.com/skumar1996github/masters-thesis-sagar-kumar>

The original data for both Yelp and Amazon can be found in the references section at the end listed at [4] and [5] respectively.

1. Introduction

1.1 Background and Problem Statement

Customer reviews and opinions have a significant impact on business operations and strategies. And the sudden and rapid growth of e-commerce platforms like Yelp and Amazon have also changed the ways that consumers purchase and experience products and services. Also, as the globe becomes more and more digitized and online based, the consumers rely heavily on online reviews when making a purchasing decision. Therefore, online ratings/stars and written reviews have become important sources of information that help consumers evaluate the quality of the product or service and get a better idea about the business reputation and past customer satisfaction before ultimately making a purchase. This thesis will explore the relationship and impact of online customer reviews and sales performances of both online and offline products/services. [1]

We will particularly be looking at two different companies, one which deals with local brick-and-mortar businesses such as restaurants, cafes, retail stores etc. and one which deals only with e-commerce products and services which are strictly sold online. More specifically, in our case for brick-and-mortars stores we will review business review data from **Yelp**, an online platform that allows customers to review and rate local businesses in their area and for online products we will analyze data from **Amazon**, one of the largest e-commerce platforms for online retail products.

Overall, the aim of this thesis is to analyze the impact of online reviews and customer review sentiment on consumer demand and business revenue. The results of this thesis should particularly be useful for companies that not only collect online reviews but also rely heavily on them to make future business decisions on performance and customer satisfaction, two factors that heavily influence business revenue and growth potential. In addition, the findings from this thesis may help businesses better understand how online ratings and customer sentiment affect consumer behavior and demand as well as overall market competitiveness. [1]

1.2 Online reviews and ratings systems

An online review generally allows a customer to voice their opinion about their experience doing business with a company. It usually consists of allowing a customer to write a written text review, along with attached images as an option. Additionally, one of the most common and quick ways to leave a review is by the star rating. The range varies depending on the platform, usually 1 thorough 5 or 1 through 10, where 1 is the lowest rating (customer is dissatisfied) and 10 is the highest rating (customer is satisfied). Higher ratings usually are associated with better customer satisfaction and better business reputation. Another important concept is the review sentiment. It refers to the emotional tone of the opinion expressed in the written review. Values lie within the range of positive (+1), neutral (0) and negative (-1) tones and they provide a deeper understanding of the free-text review along with the star rating.

1.3 Measuring Consumer Demand

Consumer demand as a metric can be tough to measure, especially when data is usually incomplete or unauthentic. Not to mention, most companies do not reveal their sales and revenue data due to various reasons like privacy concerns, maintaining a competitive advantage among many others. Therefore, for our thesis, we will measure the number of reviews as a proxy for consumer demand for any product or business. Meaning, the amount of engagement a product or business receives will be like an indicator of its overall popularity and demand within the consumer market.

1.4 Structure of the thesis

In order to tackle the question of how online reviews influence consumer demand, this thesis applies multiple data analysis and machine learning techniques to real-world datasets containing real and authentic customer and business review information from Yelp and Amazon. After importing and cleaning the dataset, we will conduct an initial data exploration which includes looking at rating and review length distributions, average review counts amongst different rating levels and more. Afterwards, we will conduct a sentiment analysis which involves calculating a sentiment score for each text review to determine its sentiment polarity and how it relates to customer ratings and overall consumer demand. Then we will conduct an OLS regression analysis to look at the relationship between different variables within our dataset and how much they contribute to a product or business demand. Lastly, we will use machine learning techniques to predict consumer demand by using features like business ratings, review sentiment, review length, business location information amongst others. More precisely, we will be using a Random Forrest model to perform our predictions. In the end, we will look at a few limitations of the thesis itself as well as discuss future works.

2. Literature Review

In order to better understand the relationship between online reviews and consumer demand, it is important to look at previous academic research done by researchers in a similar area. This will allow us to compare findings and maybe even identify common patterns as well as similar limitations they might have run into.

The first paper we will look at is by Nan et. al. (2008) titled *“Do Online Reviews Affect Product Sales? The Role of Reviewer Characteristics and Temporal Effects.”*. The paper looks at how online reviews affect product sales and consumer purchasing behavior. The so-called reviewer characteristics from the title refers to aspects about the review like review scores, reviewer reputation, reviewer exposure, and the timing the review was made. This is important because this is how this study differs from ours as they are using properties about the review to get a better understanding of its influence. Our study plans to use review sentiment, review length and star ratings as characteristics used to analyze the

relationship with consumer demand. The study found that customers pay more attention to the credibility of reviewers rather than only paying attention to star ratings. One interesting thing the study found was that the reviews have a much less effect on consumer behavior after the product has been released for a while. [2]

The second paper we will look at is by Oktadinna et. al. (2024) titled *“The Impact of Online Reviews and Ratings on Consumer Purchasing Decisions on E-commerce Platforms”*. It investigates how online reviews and ratings affect consumer purchasing decisions on e-commerce platforms like Tokopedia, Shopee and Bukalapak. The study’s methodology was to collect data through online surveys of 180 active e-commerce consumers. The data was then analyzed with descriptive statistics, Pearson correlation tests and more. The study found that positive reviews and high product ratings significantly increase consumer trust and purchase intentions and that negative reviews reduce the likelihood of purchasing a product. Therefore, we can probably also expect in our thesis study the same results. [3]

3. Data Overview

3.1 Yelp Data Overview

Data for Yelp was collected from their official website which can we found [here](#), as well as in the references. The dataset is divided in two-fold. The first part is the business review data itself. It contains information about the business, namely business id, name, its location (state and city within the United States), review count, business attributes and categories. The second half consists of anonymized review data for specific businesses which has the review id, which user wrote the review and to which business, how many stars they gave, the review itself and the date it was written. Using the business ID attribute, we were able to combine the two datasets such that we have one data frame which consists of reviews for each business, its ratings and review count. Lastly, the dataset includes two types of ratings, the first rating is per review, and the second rating is the rating of the business as a whole which is the average of all reviews the business receives. For the purposes of this thesis, we will consider only the average business ratings as it already aggregates the data per business. [4]

3.2 Amazon Data Overview

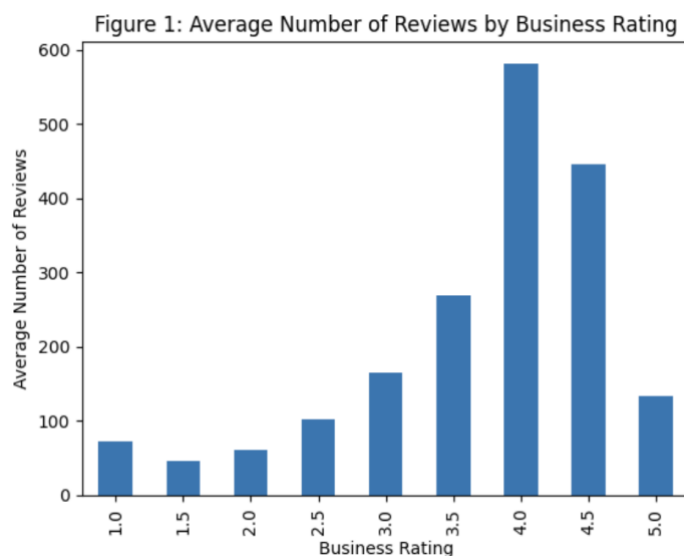
Data for Amazon was collected from the Amazon Review Data 2018 which is provided by Jianmo Ni from the University of California San Diego. It can be found [here](#) as well as in the references. It consists of large-scale real-world anonymized product review information about electronics sold on Amazon Marketplace. The data source is divided into two halves, the first half contains data about the product review including their rating, review text, date and time it was written and more. The second half consists of information about the product itself, which was reviewed including its category, description, brand,

price and much more. We combined the data based on the product ID (ASIN) to create one final data frame consisting of review information per product. [5]

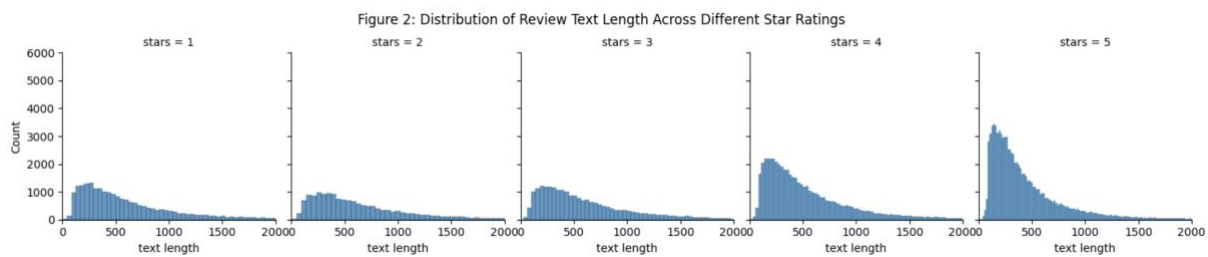
4. Exploratory Data Analysis

4.1 Exploring Yelp's Dataset

We will start our exploratory data analysis by looking at Yelp's distribution of business ratings and review counts in order to get a better understanding of the overall structure of the dataset. We achieve this by visualizing star rating frequencies and average review counts across rating levels. We can see from Figure 1 that businesses that have higher average ratings tend to have a larger number of reviews than those with lower average ratings. This suggests a positive correlation between business reputation and consumer demand – because in our case the number of reviews is used as a proxy for consumer demand. In particular, we see that businesses with ratings between 4 and 4.5 seem to have the greatest number of reviews which shows stronger customer engagement, popularity and also builds trust.

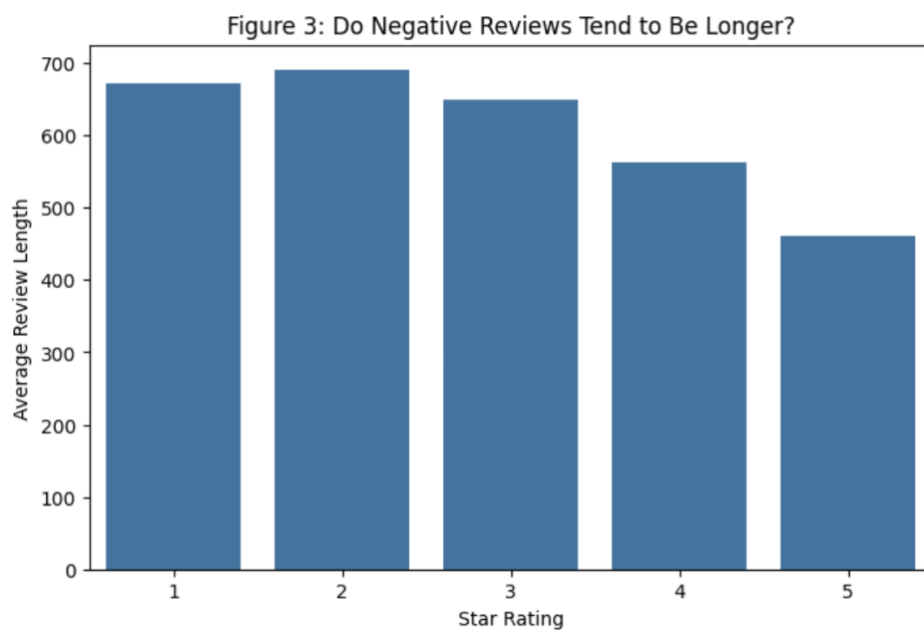


Now we want to look at the distribution of review text lengths across the five different star rating levels. We have depicted this in Figure 2 and we can observe that higher-rated reviews, particularly 4-star and 5-star reviews, appear significantly more frequently throughout the dataset compared to lower-rated reviews. Also, lower-rated reviews tend to contain longer written feedback on average, probably because dissatisfied customers may be more likely to provide detailed explanations of their experiences.

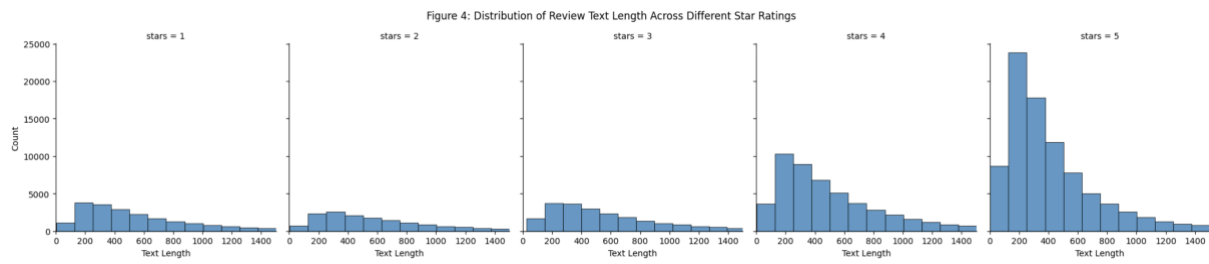


4.2 Exploring Amazon's Dataset

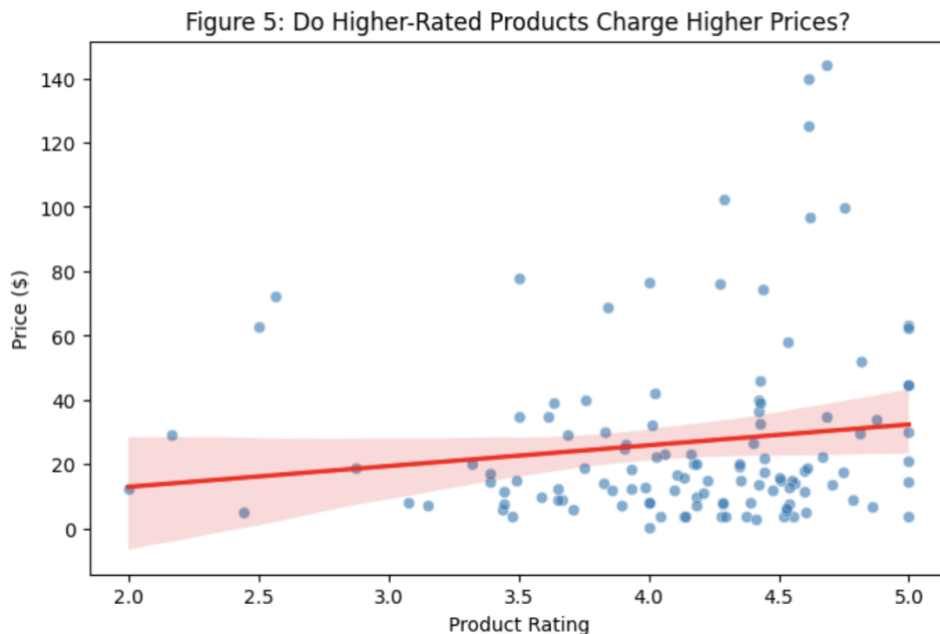
To explore the amazon review dataset, we are firstly looking at the relationship between product ratings and average review length of that product. From Figure 3, we observe that lower-rated products at around ratings of 1 to 2 have much longer reviews on average compared to higher rated products. This suggests that customers that are dissatisfied with the amazon product usually write much longer and detailed responses which convey their experience to other users. Whereas satisfied customers leave much shorter and to the point reviews showing their satisfaction with much less words.



We will also be looking at the distribution of review text length across different products ratings similar to what we did with Yelp. Figure 4 shows us that 4-star and 5-star reviews appear much more frequently within our dataset compared to lower rated star reviews. Also note that most reviews are relatively short while long reviews occur less frequently. Overall, this we can see that customers are generally more likely to leave positive reviews, while only a smaller proportion of users provide longer and detailed written feedback.



We also looked at the relationship between the products ratings versus its price, which has been depicted in Figure 5 below. We observe a slight positive relationship between product ratings and product prices, as indicated by the upward-sloping regression line. However, this relationship is not too strong as the slope of the line is not too steep and also there is a huge amount of variance in prices across all levels. This suggests that while product ratings may influence pricing to some extent, other factors likely also play a significant role in determining product prices – which we will now investigate.

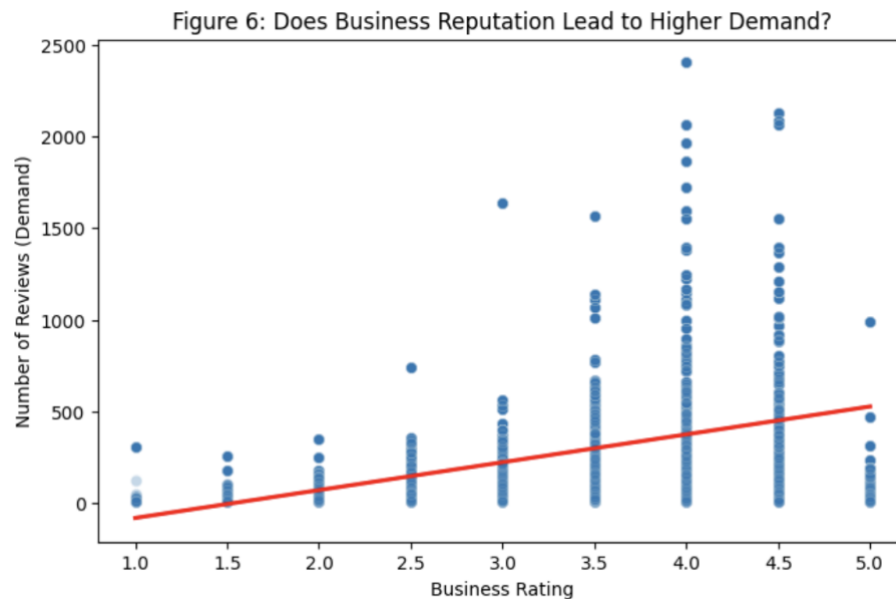


5. Ratings vs. Demand

5.1 Yelp

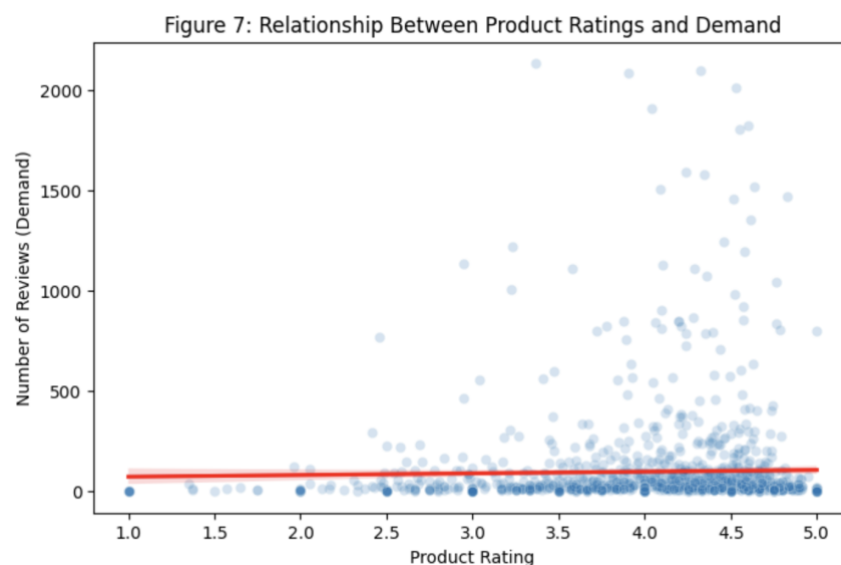
In order to understand the relationship between ratings and demand (number of reviews), we created a scatterplot comparing business ratings against review counts as well as a regression line to get an idea of the general trend between the two factors. We observe from Figure 6 that there seems to be a slightly positive relationship between the two. Meaning, as business ratings get higher, they generally tend to receive more reviews and hence are more popular. Especially businesses that lie within the ratings range of 3.5 and 4.5, they appear to have the highest level of engagement within our dataset. Overall, the

main take-away from this visualization is that higher business ratings may contribute to greater consumer demand and popularity, as businesses with stronger customer satisfaction levels generally tend to attract more customer engagement and review activity.



5.2 Amazon

We have applied the same methodology as we did with Yelp to the Amazon dataset for understanding the relationship ratings and demand. However, for Amazon, we are comparing products ratings rather than business ratings and the ratings are continuous in nature rather than fixed into denominations of 0.5. We observe that there appears to be a very slight positive relationship between product ratings and customer demand. However, this relationship is quite weak due to the large spread of data points throughout the dataset that dampen the effect or correlation. Overall, this visualization tells us that even though we can say that higher ratings lead to more demand, there are still factors outside of the dataset that also heavily factor in when calculating demand.

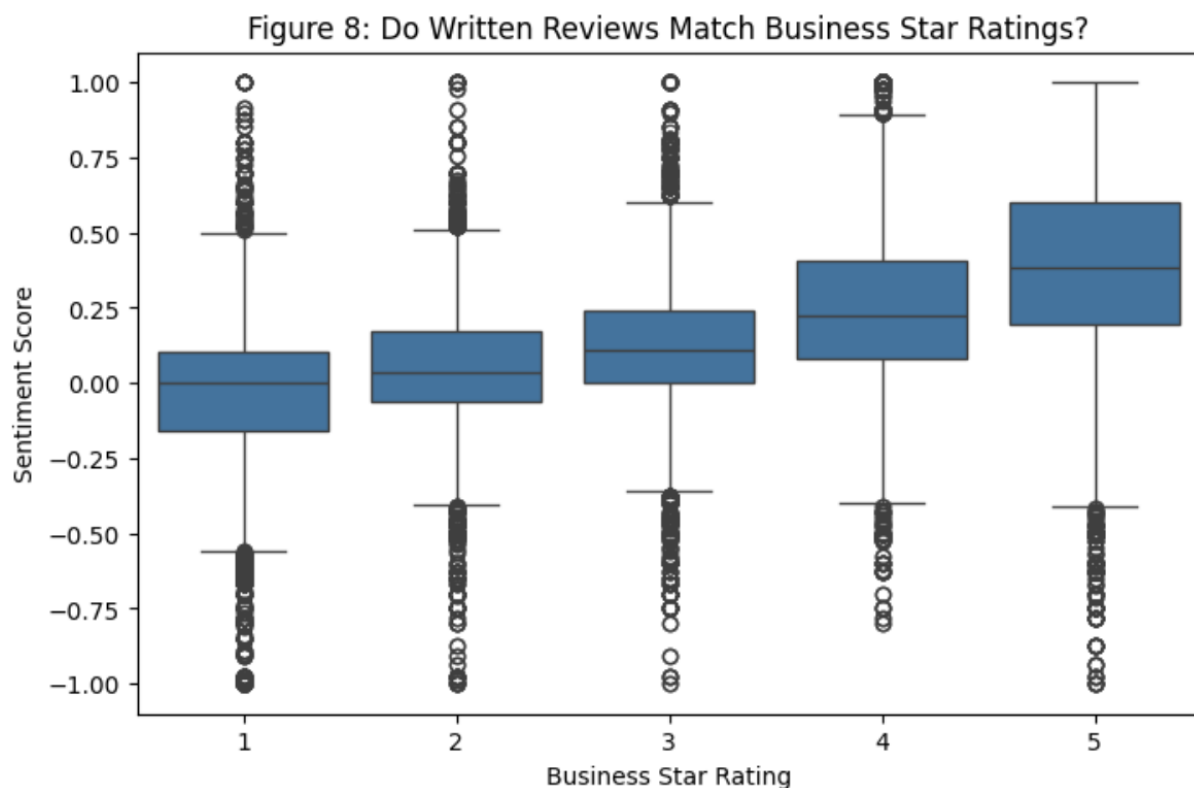


6. Sentiment Analysis

In order to better understand the emotional tone of the opinions expressed in the customer reviews, we also conduct a sentiment analysis for the review texts from both Yelp and Amazon. In general, Sentiment analysis allows us to confirm whether the written content of customer reviews aligns with the numerical star ratings provided by users. Therefore, by performing this type of analysis we can help validate the consistency and reliability of the dataset by examining whether the written review sentiment aligns with the numerical ratings provided by customers. Moreover, calculating a sentiment score for each review will also help us later on as we use the sentiment score as an independent variable within our regression and machine learning models to analyze its relationship with consumer demand and business popularity. [6]

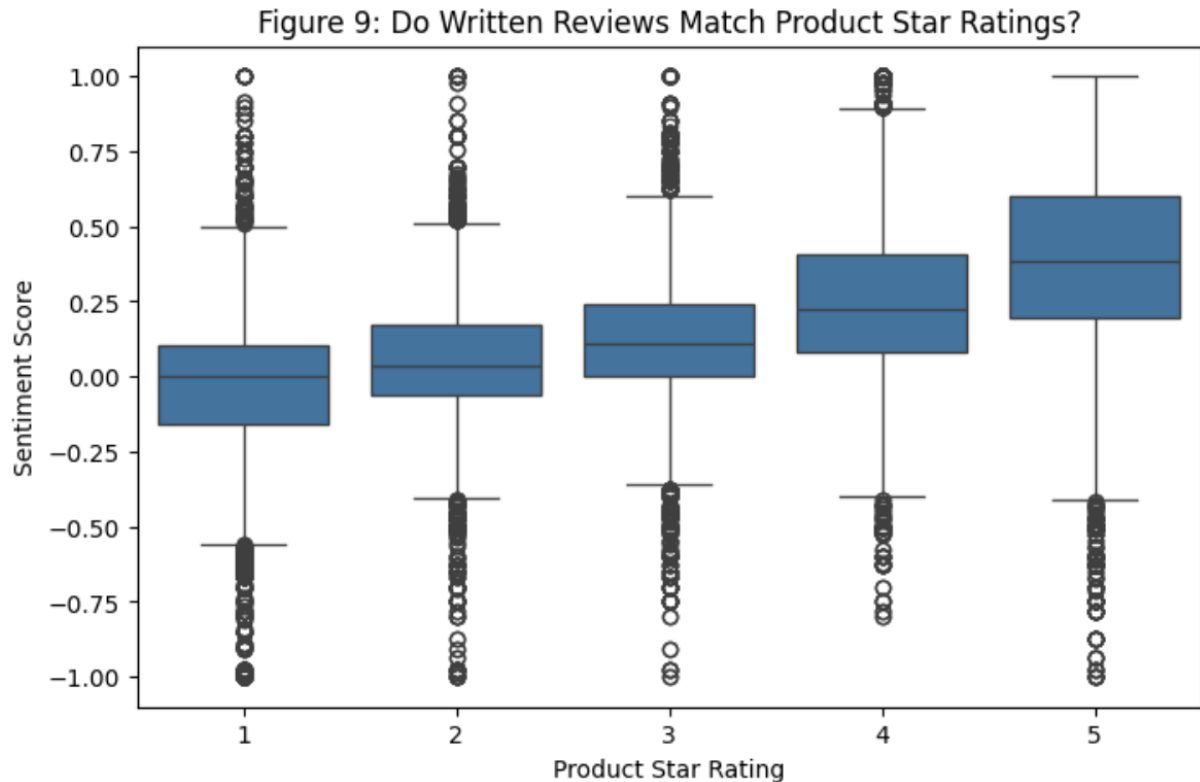
6.1 Yelp Dataset Sentiment Analysis

Looking below at the sentiment analysis of our Yelp dataset, we can observe that sentiment scores generally increase as business star ratings become higher. Reviews associated with 1-star businesses tend to have more negative to neutral sentiment scores, while reviews associated with 4-star and 5-star businesses show significantly more positive sentiment values. This is a good indicator because this suggests that the written review content largely aligns with the numerical ratings provided by customers, which also helps us to validate the consistency and reliability of the dataset.



6.2 Amazon Dataset Sentiment Analysis

Comparing the sentiment analysis results from Yelp to Amazon, we see that a very similar pattern exists within both datasets. As product star ratings increase, the average sentiment scores of the written reviews also become increasingly positive. Lower-rated products tend to have more negative sentiment values and highly rated products are associated with strongly positive review sentiment. This once again supports the validity and consistency of the datasets, as the written customer feedback aligns closely with the numerical star ratings provided by users and hence are not random.



7. OLS Regression Analysis

7.1 Methodology

As the next step of our thesis, we will perform an OLS regression analysis. OLS stands for Ordinary Least Squares, which is a statistical method used to examine the relationship between one dependent variable and multiple independent variables. The reason we picked OLS specifically is because it allows us to statistically measure and quantify the relationship between our variables in a simple and interpretable way. In the context of this thesis, OLS regression allows us to measure how factors such as business ratings, review sentiment and review length ultimately influence consumer demand. By performing this analysis, we can better understand which variables have the strongest statistical relationship with demand and whether these relationships are positive or

negative. In order to analyze the results in a meaningful manner, we will mainly be looking at the regression coefficients, p-values, R-squared values and model significance. [7]

7.1 Results Analysis

For our OLS regression model for the Yelp and Amazon dataset, we have decided to use **star ratings**, review **sentiment**, and review **length** as independent variables in order to analyze their relationship with consumer demand and business popularity, our dependent variable. Let's break down the results.

The first results we want to focus on is the R-squared value. It represents how much variation in the dependent variable can be explained by our chosen independent variables. Ideally, we are looking for a relatively high R-squared value because it shows that the model explains a larger proportion of variation for the dependent variable. Values range between 0 and 1, where 1 explains all the variance in the dataset. For the Yelp dataset, we got a R-squared value of 0.074, meaning that approximately **7.4%** of the variation in consumer demand can be explained by the independent variables. Similarly with Amazon, which yielded a value of 0.015 or **1.5%**, which means that the selected review-related variables have only little explanatory power in predicting consumer demand for both datasets and that factors such as ratings, sentiment, and review characteristics do influence demand to some extent but demand is heavily affected by many additional external factors not included in our model. Secondly, we want to pay attention to the regression coefficients and p-values for our independent variables. Coefficients help us to understand whether the relationship between our variables is negative, positive or neutral, whereas p-values tell us whether the variables are statistically significant or random chance. For Yelp, our coefficient values are as follows: star rating is 0.5249, sentiment is 0.3623 and review length is almost 0. The p-value for all three values is also almost 0, which means that all three variables are statistically significant in determining demand. Also, all coefficients are positive which means that more of each variable (longer reviews, higher ratings and positive sentiment) leads to more demand.

For Amazon, the coefficients and p-values are as follows: star rating is 0.0240 (p-value of 0.724), sentiment is -0.6438 (p-value of 0.157) and review length is -0.0006 (p-value of 0). The negative coefficient values suggest that higher sentiment scores and longer review lengths are associated with slightly lower consumer demand within the Amazon dataset. Whereas the p-values suggest that only review length is statistically significant within the Amazon regression model, while the rest do not have a meaningful statistical relationship with consumer demand. To summarize, the OLS regression results indicate that online review characteristics have a stronger and more statistically significant relationship with consumer demand for Yelp businesses compared to Amazon products. Whereas the Amazon results were considerably weaker and less statistically significant overall.

8. Machine Learning Predictions

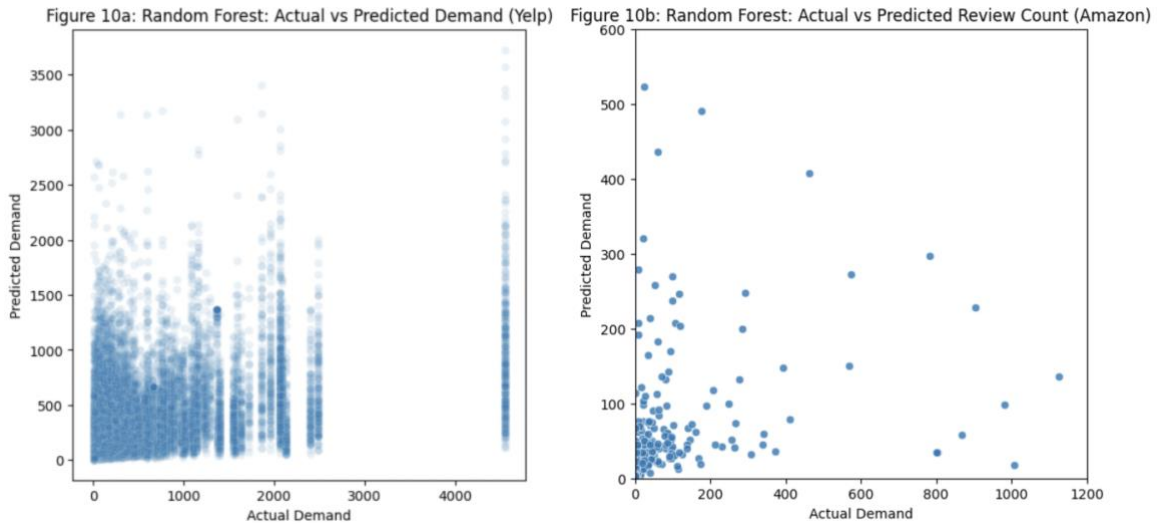
For our machine learning predictions, we will be using the Random Forest Regressor model to predict consumer demand (i.e. number of reviews) using review length, star rating and sentiment. For Yelp, we also have available to us the city and state which the business is located in so we will also be including that for prediction purposes. We chose the Random Forest Regressor model because it is capable of handling complex and non-linear relationships between variables while also being relatively robust against overfitting and outliers. [8]

8.1 Root Mean Squared Error

The first value we calculate from our model is the root mean squared error (RMSE) value which represents the average difference between the predicted values and the actual observed values. For Yelp businesses, we got a value of 561.7, and for Amazon products we got a value of 287.5 for RMSE. So, for example, Yelp model's predicted review counts differ from the actual values by approximately 561 reviews. Since the RMSE value for Amazon is lower, it suggests that the Random Forest model produced slightly more accurate predictions for the Amazon dataset compared to Yelp. This could be due to many reasons like difference in dataset structure, customer behavior of online and offline products and more.

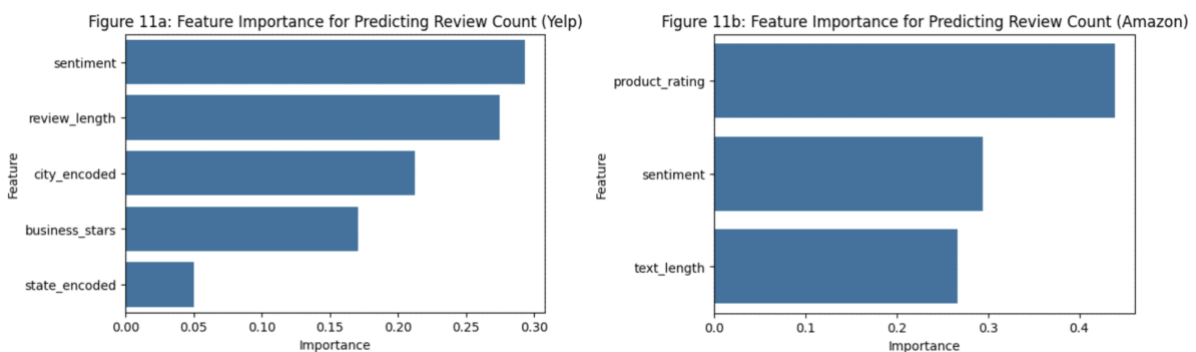
8.2 Actual vs. Predicted Review Count

Figure 10a and 10b depict the relationship between the predicted review counts and the actual review counts for Yelp and Amazon respectively. Looking at Yelp first, we can see that data points are scattered throughout the chart. This highlights that predicting the exact number of business demand remains difficult even though the model in general is capable of identifying general trends. One reason for this can be that demand is affected by external factors not included in the dataset like business quality, competition, marketing strategy, pricing strategy and more. Comparing this with Amazon's visualization, we can see that the data points are relatively more concentrated compared to Yelp. Having said that, there is still some scattering happening here as well. This model is far more accurate than the Yelp which also explains the lower RMSE value. However, because there is still scattering, it means there are factors missing in the dataset which help explain consumer demand.



8.3 Feature Importance

The last part of our prediction model focuses on feature importance. It helps us to better understand which of our chosen independent variables contribute most to predicting customer demand. The proportion of all combined variables is 100% (or 1). For Yelp, we observe that sentiment contributes the most towards predicting customer demand at 29% with review length a close second at 27%. For Yelp businesses, it seems that the state it is located in contributes the least to how much demand a business can expect. As for amazon products, its rating has the most to do with how popular or in demand a product is at roughly 44% with sentiment and review length at 29% and 26% respectively. This is because all three variables play almost an equal role in determining the demand. Whereas with Yelp businesses, sentiment and review length account for 56% or more than half in terms of predicting demand.



9. Limitations and Future Research

9.1 Review count as proxy for customer demand

A major limitation of our thesis is that we measured the number of reviews as a proxy for customer demand rather than sales data or revenue data. Although in theory number of reviews does indicate the popularity of an item as we have seen through our results, it does not tell the full story or represent demand on a 1 to 1 level. For example, many customers buy products without leaving any reviews. In addition, it may be possible that a restaurant undergoes through controversy and is flooded by negative comments. Therefore, for the purposes of our thesis, review count should be interpreted as an indirect approximation of consumer demand rather than a precise measurement.

9.2 Fake and Biased Reviews

A notable limitation for our thesis is that it is possible that Yelp businesses or Amazon product sellers may create fake reviews and ratings that are either paid beforehand to give a review or are from the sellers themselves. They do this in order to create a fake sense of trust and reputation within the market. Therefore, as a result some datasets may not be entirely accurate or trustworthy. Although this is an issue that is larger at hand and cannot be avoided within the scope of the thesis however it is important to note that it does exist and can slightly skew our results with bias. [9]

9.3 Sample Size of Dataset

A limitation of our datasets is that both the Yelp and Amazon analyses required combining multiple tables in order to create the final datasets used for our models. So, review data had to be merged with business or product metadata using identifiers such as business IDs or product ASINs. As a result, only data points that existed within both tables could be included in the final merged dataset. Moreover, we did not use the entire dataset as it was too large (upwards of 7 gigabytes) and processing times were too long. Therefore, only portions of the datasets were used for practical reasons, which means the final results may not fully represent the entire Yelp and Amazon datasets.

9.4 Future Research

For future research related to this project, we could start with first improving on the limitations. For example, to tackle the limitation of fake reviews and biases we could review some kind of fake review detection AI algorithm that based on the reviews meta data can automatically flag a review as fake or biased and then we can act accordingly in the analyzation. Secondly, to address the issue of review count as proxy for demand count, we can rather use real-world sales and revenue in combination with our existing data to get a cleaner picture on how reviews affect consumer demand. Outside of the limitations, an improve we can make to our project is to incorporate additional variables about business or products that help us improve our picture of the relationship between online reviews and consumer demand. For example, for businesses we could have used variables like

opening hours, pricing strategy, marketing strategy, brand reputation values, and their level of service provided. On the other hand, for amazon products, we could have used variables like brand popularity, advertising exposure, any potential discount strategy or even shipping speed.

10. Conclusion

This thesis aimed to investigate the relationship between the nature of online reviews of a business and products to its consumer demand. We collected and analyzed real-world anonymized data from Yelp and Amazon and investigated factors like star ratings of a review, the length of the review and its sentiment and how they relate to consumer demand.

In order to achieve this, we first did some initial exploratory data analysis just to get an idea about the structure of the dataset. For both Yelp and Amazon, we observed that businesses/products with higher average ratings generally received a greater number of reviews, suggesting a positive relationship between customer satisfaction and consumer demand. We also found that lower-rated reviews were longer and more detailed for both platforms, while higher-rated reviews appeared more frequently throughout the dataset. Lastly, for Amazon, we observed a slight positive relationship between product ratings and product prices, although we noted that external factors likely also contribute to determining product prices and consumer demand.

When it comes to investigating the relationship between reviews and demand, in this thesis, we used the number of reviews a business/product gets as a proxy for its popularity and demand. Based on this, for both Yelp and Amazon we found that both datasets showed a positive relationship between higher ratings and greater consumer demand, meaning that businesses or products with better ratings generally tend to receive more reviews and customer engagement. However, for both cases the relationship was not perfectly strong or linear, suggesting that additional external factors also influence consumer demand and popularity. The main difference between the two platforms was that the relationship between ratings and demand was stronger for Yelp than for Amazon. For Amazon, the relationship was weaker because many other factors such as price, brand reputation, advertising strategy and more play a part in determining its demand.

As the next step, we decided to do a sentiment analysis for the written reviews of both datasets. We did a sentiment analysis because it allows us to understand the emotional context of the customers opinion. Also, it allowed us to confirm whether the written review matches the star ratings. Meaning a low star rating should correlate with a neutral to negative sentiment, as an example. What we found was that for both Yelp and Amazon, the expected pattern occurs. Meaning, the expected sentiment matches the actual star ratings and review. This is a good sign as it verifies that the reviews are not just random data points but rather real-world opinions from customers.

We also performed ordinary least squares regression to get a deeper understanding how our independent variables like review length, review sentiment and star ratings contribute to finding consumer demand and by how much. For Yelp businesses we found that all three independent variables had a positive and significant relationship with demand. However, we got a low R-squared value of just 7.4%, which means that 7.4% of the variation in the data is explained by our variables. This is actually quite low which means that a large portion of demand is still affected by many additional external factors that were not included in our model or dataset. For Amazon we found that only review length was statistically significant within the model. Factors like review sentiment and star ratings did not play much of a role in determining demand. In addition, the R-squared value for amazon was even lower at 1.5%. This highly suggests that external factors not within our scope likely play a much larger role.

We also applied machine learning prediction models such as Random Forest in order to predict consumer demand. The model produced more accurate predictions for Amazon than for Yelp, as shown by the lower RMSE value and the more concentrated prediction results. In addition, the feature importance analysis showed that sentiment and review length were particularly important for predicting Yelp business demand, while product ratings played the largest role for Amazon products.

In the end we also looked at some limitations of our thesis project and discussed future works. These limitations included the possibility of fake and biased reviews forged by the business or product owners to fake a good business reputation and show demand to other customers. We also discussed the limitations of using number of reviews as a proxy for consumer demand as it is not a 1 to 1 relation and many exceptions exist like for example a company being flooded by negative comments after undergoing controversy. For future work, we suggested using real-world sales or revenue data or even incorporating additional business and product variables and applying fake review detection methods in order to improve the accuracy and depth of our analysis.

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